

Chapter 9 Forms

Overview of Forms

Forms are used
all over the Web to

- Accept information
- Provide interactivity

Types of forms:

- Search form, Order form, Newsletter sign-up form, Survey form, Add to Cart form, and so on...

Form

- An HTML element that contains and organizes form controls such as text boxes, check boxes, and buttons
- that can accept information from website visitors.



Figure 9.1 This form accepts shipping information

Two Components of Using Forms

1. The HTML form - the web page user interface and
2. The server-side processing - Server-side processing works with the form data and sends e-mail, writes to a text file, updates a database, or performs some other type of processing on the server.

HTML Using Forms

<form>

- Contains the form elements on a web page
- Container tag

<input>

- Configures a variety of form elements including text boxes, radio buttons, check boxes, and buttons
- Stand alone tag

<textarea>

- Configures a scrolling text box
- Container tag

<select>

- Configures a select box (drop down list)
- Container tag

<option>

- Configures an option in the select box
- Container tag

Sample Form HTML_(1 of 2)

<form>

```
E-mail: <input type="text" name="email" id="email" required><br>  
<input type="submit">  
<input type="reset">
```

</form>



Figure 9.3 The browser displayed an error message

HTML form element

- The form element attributes:
 - action
 - Specifies the server-side program or script that will process your form data
 - method
 - get – default value, form data passed in URL
 - post – more secure, form data passed in HTTP Entity Body
 - name
 - Identifies the form
 - id
 - Identifies the form

Input Text box

<input>

Accepts text information

- Attributes:
 - type="text"
 - name
 - id
 - size
 - maxlength
 - value

Sample Text Box

E-mail:

Figure 9.2 The <input> tag with type="text" configures this form element

textarea Scrolling Text Box

`<textarea>` `</textarea>`

Configures a scrolling text box

Attributes:

- name
- id
- cols
- rows

Sample Scrolling Text Box

Comments:



Figure 9.9 Scrolling text box

input Submit Button

`<input>`

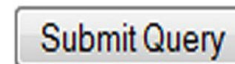
Submits the form information When clicked:

- Triggers the **action** method on the `<form>` tag
- Sends the form data (the name=value pair for each form element) to the web server.

Attributes:

- type="submit"
- name
- id
- value

Sample Submit Button



input Reset Button

`<input>`

Resets the form fields to their initial values

Attributes:

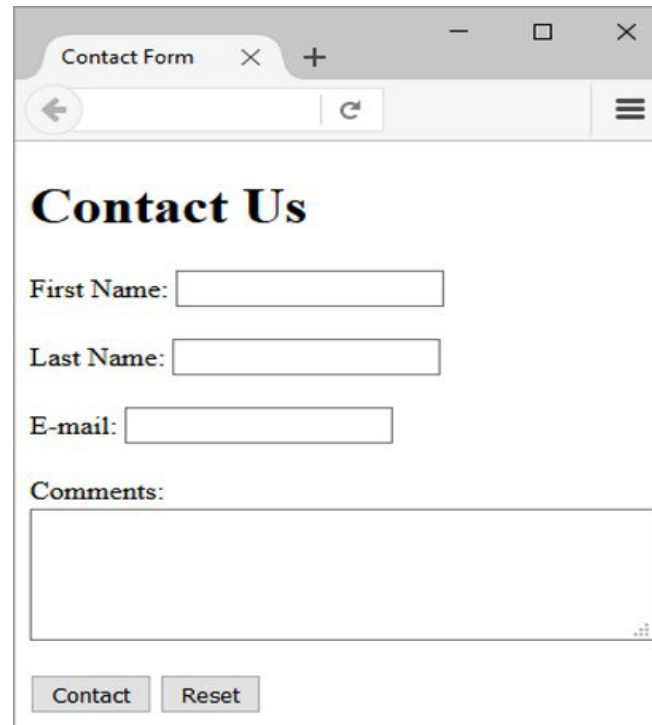
- type="reset"
- name
- id
- value

Sample Reset Button



Hands-On Practice: You can try developing a form as shown below

Figure 9.10 A typical contact form



The image shows a web browser window with a single tab titled 'Contact Form'. The browser's address bar is empty, and the page content is a contact form. The form has a title 'Contact Us' in a large, bold, black serif font. Below the title are four input fields: 'First Name:', 'Last Name:', 'E-mail:', and 'Comments:'. The 'First Name:', 'Last Name:', and 'E-mail:' fields are single-line text boxes. The 'Comments:' field is a larger, multi-line text area. At the bottom of the form are two buttons: 'Contact' and 'Reset'. The browser window has standard window controls (minimize, maximize, close) in the top right corner.

Contact Form

← | →

Contact Us

First Name:

Last Name:

E-mail:

Comments:

Contact Reset

input File Upload Control

<input>

Accepts a file upload when the form is submitted

Requires the form element to be configured with method="post", enctype="multipart/form-data", and the action attribute.

- Attributes:
 - type="file"
 - name
 - id
 - accepts
 - capture
 - multiple

Sample File Upload Control

File Upload: No file selected.

input Password box

<input>

Accepts text information that needs to be hidden as it is entered

- Attributes:
 - type="password"
 - name
 - id
 - size

Sample Password Box

Password:

Figure 9.8 The characters secret9 were typed, but the browser does not display them. (Note: Your browser may use a different symbol, such as a stylized circle, to hide the characters.)

input Check box

<input>

Allows the user to select one or more of a group of predetermined items

- Attributes:
 - type="checkbox"
 - name
 - id
 - checked
 - value

Sample Check Box

Choose the browsers you use:

- ☐ Google Chrome
- ☐ Firefox
- ☐ Microsoft Edge

Figure 9.6 Sample check box

input Radio Button

<input>

Allows the user to select exactly one from a group of predetermined items

Each radio button in a group is given the same name and a unique value

- Attributes:
 - type="radio"
 - name
 - id
 - checked
 - value

input Hidden form data

<input>

This form control is *not* displayed on the web page.

Hidden form fields

- Can be accessed by both client-side and server-side scripting
- Sometimes used to contain information needed as the visitor moves from page to page.

Attributes:

- type="hidden"
- name
- id
- value

Sample Radio Button

Select your favorite browser:

- ☐ Google Chrome
- ☐ Firefox
- ☐ Microsoft Edge

Figure 9.7 Use radio buttons when only one choice is an appropriate response

Select List

`<select></select>`

Configures a select list (along with option elements)

Also known as: Select Box, Drop-Down List, Drop-Down Box, and Option Box.

Allows the user to select one or more items from a list of predetermined choices.

Attributes:

- name
- id
- size
- multiple

Select List: One Initial Visible Item

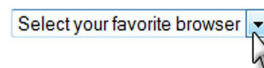


Figure 9.11 A select list with size set to 1 functions as a drop-down box when the arrow is clicked

Options in a Select List

`<option></option>`

Configures the options in a Select List

Attributes:

- value
- selected

Select List: Four Items Visible



Figure 9.12 Because there are more than four choices, the browser displays a scroll bar

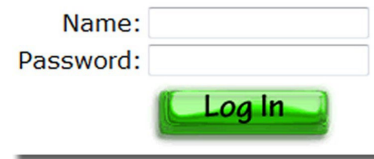
input Image Button

<input>

Submits the form

When clicked:

- Triggers the **action** method on the form tag
- Sends the form data (the name=value pair for each form element) to the web server.
- Attributes:
 - type="image"
 - name
 - id
 - src



Name:

Password:

Figure 9.13 The web page visitor will select the image button to submit the form

Button Element

<button></button>

A container tag

When clicked, its function depends on the value of the type attribute.

Can contain a combination of text, images, and media

- Attributes:
 - type="submit", type="reset", type="button"
 - name
 - id
 - alt
 - value



Name:

E-mail:

Sign up for free newsletter

Figure 9.14 The button element configured as a submit button

Label Element

`<label></label>`

Associates a text label with a form control Two Different Formats:

`<label>Email: <input type="text" name="email" id="email"></label>`

Or

`<label for="email">Email: </label>`

`<input type="text" name="CustEmail" id="email">`

Fieldset and Legend Elements

The Fieldset Element

- Container tag
- Creates a visual group of form elements on a web page

The Legend Element

- Container tag
- Creates a text label within the fieldset

Fieldset and Legend Elements

```
<fieldset><legend>Customer Information</legend>
  <label>Name:
    <input type="text" name="Name" id="Name"></label>
  <br><br>
  <label>Email:
    <input type="text" name="Email" id="Email"></label>
</fieldset>
```

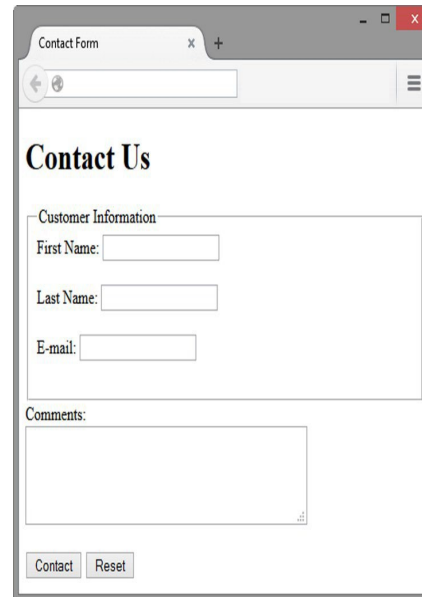


Figure 9.16 The fieldset, legend, and label elements

tabindex attribute

Attribute that can be used on form controls and anchor tags
Modifies the default tab order

Assign a numeric value

```
<input type="text" name="CustEmail" id="CustEmail" tabindex="1">
```

Accesskey Attribute

Attribute that can be used on form controls and anchor tags
Create a “hot-key” combination to place the focus on the component
Assign a value of a keyboard letter

On Windows use the CTRL and the “hot-key” to move the cursor

```
<input type="text" name="CustEmail" id="CustEmail" accesskey="E">
```

Using CSS Float to Style a Form

```
form { background-color: #EAEAEA; font-family: Arial, sans-serif;
```

```
width: 350px; padding: 10px;}
```

```
label { float: left;
```

```
clear: left; display: block;
```

```
width: 100px;
```

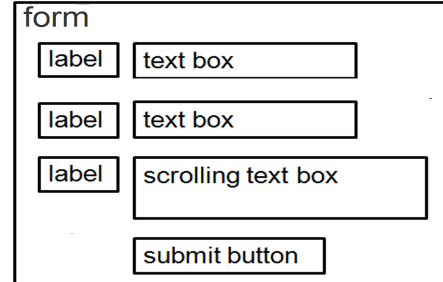


Figure 9.18 Wireframe for a form

Using CSS Float to Style a Form

```
text-align: right; padding-right: 10px; margin-top: 10px; }
```

```
input, textarea { margin-top: 10px;  
                  display: block;}
```

```
input[type="submit"] { margin-left: 110px; }
```

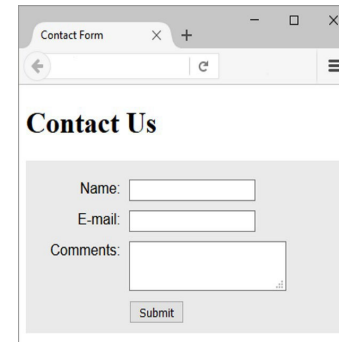
A screenshot of a web browser displaying a 'Contact Us' form. The browser window has a title bar with 'Contact Form' and standard navigation buttons (back, forward, refresh, home). The form itself is titled 'Contact Us' in a bold, black font. Below the title, there is a light gray rectangular area containing the form fields. The fields are labeled 'Name:', 'E-mail:', and 'Comments:'. The 'Name' and 'E-mail' fields are single-line text inputs, while the 'Comments' field is a multi-line text area. A 'Submit' button is located at the bottom right of the form area. The form is styled with CSS, including margins and padding.

Figure 9.19 This form is configured with CSS